Topic 3: Analysis under the Completely Randomized Design

Chapters 11.1-11.4 and 15.6 of our Textbook

STAT 2509: Introduction to Statistical Modeling II

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Learning Objectives

By the end of this lecture, you should be able to:

- Understand basic terminology in experimental design
- Analyze completely randomized designs using ANOVA
- Test assumptions of ANOVA including homogeneity of variances
- Perform multiple comparisons using Tukey's HSD
- Apply nonparametric alternatives: Kruskal-Wallis and Dunn's tests
- Interpret statistical output in the context of real problems

Basic Terminology in Experimental Design

Term	Definition and Example
Factor	A categorical explanatory variable being manipulated. *Example:* Fertilizer type, Drug dosage, Teaching method
Level	A specific value/category of a factor. *Example:* For factor "Fertilizer": Organic, Chemical, None
Treatment	A specific combination of factor levels applied to experimental units. *Example:* "Apply Organic fertilizer at planting"
Experimental Unit	A smallest unit receiving treatment independently. *Example:* Individual plant, plot of land, patient
Response Variable	A quantitative outcome being measured. *Example:* Plant height (cm), Test score, Recovery time (days)

Example

Research question: Does study method affect exam scores?
Factor: Study method.
Levels: Reading only, Practice problems, Group study.
Treatment: "Practice problems" group.
Experimental unit: Individual student.
Response: Final exam score $(0 - 100)$.

Completely Randomized Design: Detailed Notation

Design characteristics

- k treatments/groups (levels of a single factor).
- Experimental units **randomly assigned** to treatments.
- Each unit receives exactly one treatment.
- Simplest and most common experimental design.

Mathematical notation

k : Number of treatments (groups), $\quad i = 1, 2, \ldots, k$

μ_i : Population mean for treatment i,

n_i : Sample size for treatment i,

$n = \sum_{i=1}^{k} n_i$: Total sample size,

y_{ij} : Observation j in treatment i, $\quad j = 1, 2, \ldots, n_i$

$T_i = \sum_{j=1}^{n_i} y_{ij}$: Total for treatment i,

Completely Randomized Design: Detailed Notation

Mathematical notation

$$\text{G.T.} = \sum_{i=1}^{k} T_i = \sum_{i=1}^{k} \sum_{j=1}^{n_i} y_{ij} \text{ : Grand total of samples,}$$

$$\bar{y}_i = \frac{T_i}{n_i} = \frac{1}{n_i} \sum_{j=1}^{n_i} y_{ij} \text{ : Sample mean for treatment } i,$$

$$s_i^2 = \frac{\sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2}{n_i - 1} = \frac{1}{n_i - 1} \left(\sum_{j=1}^{n_i} y_{ij}^2 - \frac{1}{n_i} T_i^2 \right) \text{ : Sample variance for treatment } i,$$

ANOVA Model: Two Equivalent Forms

Form 1: Cell means model (easier for understanding)

$$y_{ij} = \mu_i + \varepsilon_{ij},$$

- y_{ij}: Response for unit j in group i.
- μ_i: True mean for group i.
- ε_{ij}: Random error $\sim N(0, \sigma^2)$ for all i, j.

Form 2: Factor effects model (better for extensions)

$$y_{ij} = \mu + \alpha_i + \varepsilon_{ij},$$

- μ: Overall grand (common) mean ($\mu = \frac{1}{k}\sum_{i=1}^{k} \mu_i$).
- α_i: Effect of treatment i ($\alpha_i = \mu_i - \mu$).
- Constraint: $\sum_{i=1}^{k} \alpha_i = 0$ (effects sum to zero).

Three critical assumptions

1. **Normality**: $\varepsilon_{ij} \sim N(0, \sigma^2)$
   Check: Normal probability plot (Q-Q plot).
2. **Independence**: Observations are independent
   Check: Experimental design, randomization.
3. **Homogeneity of Variances**: $\sigma_1^2 = \sigma_2^2 = \cdots = \sigma_k^2$
   Check: Hartley's test, Levene's test, residual plots.

Hypothesis Testing in ANOVA

Hypotheses

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k \quad \text{(All group means equal)}$$
$$H_a : \text{At least two } \mu_i \text{ are different } (\mu_i \neq \mu_j \text{ for } i \neq j)$$

Note: H_a is NOT "all means are different"!

The intuition: Comparing two sources of variation

- **Between-group variation (SSTr)**: How much group means differ from overall mean
- **Within-group variation (SSE)**: How much individuals differ from their group mean
- **Logic:** If H_0 true, these should be comparable. If H_0 false, between-group variation >> within-group

F-test statistic

$$F = \frac{\text{Variation between groups}}{\text{Variation within groups}} = \frac{\text{SSTr}/(k-1)}{\text{SSE}/(n-k)} = \frac{\text{MSTr}}{\text{MSE}}$$

Under H_0: $F \sim F(k-1, n-k)$, where $df_{SSTr} = k-1, df_{SSE} = n-k$.

We reject H_0 at significant level α if F-test $> F_{\alpha;(k-1,n-k)}$.

Sums of Squares: Detailed Calculations

Three components of variation

$$\text{TSS (Total)} = \sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{..})^2 = \sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij}^2 - \frac{(G.T.)^2}{n}$$

$$\text{SSTr (Treatments)} = \sum_{i=1}^{k} n_i(\bar{y}_i - \bar{y}_{..})^2 = \sum_{i=1}^{k}\frac{T_i^2}{n_i} - \frac{(G.T.)^2}{n} = \sum_{i=1}^{k} n_i\bar{y}_i{}^2 - n\bar{y}^2$$

$$\text{SSE (Error)} = \sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_i)^2 = \sum_{i=1}^{k}\left(\sum_{j=1}^{n_i} y_{ij} - \frac{(G.T.)^2}{n}\right)$$

$$= \sum_{i=1}^{k}(n_i - 1)S_i^2 = \text{TSS} - \text{SST}_r$$

where $\bar{y}_{..} = \frac{1}{n}\sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij} = \dfrac{1}{n}\text{G.T.}$

Note $y_{ij} - \bar{y}_{..} = (\bar{y}_i - \bar{y}_{..}) + (y_{ij} - \bar{y}_i)$. Thus, square both sides and sum to get

$$\sum_i\sum_j(y_{ij} - \bar{y}_{..})^2 = \sum_i n_i(\bar{y}_i - \bar{y}_{..})^2 + \sum_i\sum_j(y_{ij} - \bar{y}_i)^2$$

ANOVA Table: Complete Structure and Interpretation

Source	df	SS	MS	F
Treatments	$k-1$	SSTr	$\text{MSTr} = \frac{\text{SSTr}}{k-1}$	$F = \frac{\text{MSTr}}{\text{MSE}}$
Error	$n-k$	SSE	$\text{MSE} = \frac{\text{SSE}}{n-k}$	
Total	$n-1$	TSS		

Degrees of Freedom Explained

- **Treatments df =** $k-1$: With k groups, need $k-1$ comparisons to assess differences.
- **Error df =** $n-k$: Each group loses 1 df estimating its mean, total k df lost.
- **Total df =** $n-1$: Total sample loses 1 df estimating overall mean.

Expected values of mean squares (important note!)

$$E(\text{MSE}) = \sigma^2$$

$$E(\text{MSTr}) = \sigma^2 + \frac{\sum_{i=1}^{k} n_i(\mu_i - \mu)^2}{k-1}$$

- If H_0 true: $E(\text{MSTr}) = \sigma^2$ (same as MSE)
- If H_0 false: $E(\text{MSTr}) > \sigma^2$ (larger than MSE)

Hartley's F-max Test: Testing Equal Variances

When to use

- Before conducting ANOVA.
- Requires **balanced design** ($n_1 = n_2 = \cdots = n_k$).
- Assumes normality.

Hypotheses

$$H_0 : \sigma_1^2 = \sigma_2^2 = \cdots = \sigma_k^2$$

$$H_a : \text{At least two variances are different}$$

Test Procedure

1. Compute sample variances for each group: $s_1^2, s_2^2, \ldots, s_k^2$,
2. Identify maximum and minimum: $s_{\max}^2 = \max(s_i^2)$, $s_{\min}^2 = \min(s_i^2)$,
3. Compute test statistic:

$$F_{\max} = \frac{s_{\max}^2}{s_{\min}^2},$$

4. Compare to critical value: $F_{\max,\alpha}(k, [\bar{n}] - 1)$, where $[\bar{n}]$ is the integer part of $\bar{n}$, where $\bar{n} = \frac{1}{k}\sum_{i=1}^{k} n_i$ (for example $[8.5] = 8$).

Hartley's F-max Test

Critical values

- From $F_{\max}$ distribution table.
- Depends on: α, k (number of groups), $\nu = [\bar{n}] - 1$ (df within groups)

Example

$$k = 3, n_1 = 5, n_2 = 3, n_3 = 2, \alpha = 0.05, s_1^2 = 16.0, s_2^2 = 25.0, s_3^2 = 9.0$$

$$F_{\max} = 25.0/9.0 = 2.78,$$

$$[\bar{n}] = \text{integer part of } \frac{1}{3}(5 + 3 + 2) \approx 3.3$$

Thus, $\nu = [\bar{n}] - 1 = 3 - 1 = 2$

$$F_{\max,0.05}(3, 2) \approx 87.5 \qquad \text{(from Hartley's table)}$$

Since $2.78 < 87.5$, we fail to reject H_0 (variances equal).

Multiple Comparisons Problem

The Issue We're Solving

When ANOVA rejects $H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$, we need to answer:

"Which specific group means are different from each other?"

Why not just do individual t-tests for each pair?

For $k = 4$ groups, we have $m = \binom{4}{2} = \frac{4(3)}{2} = 6$ comparisons.

1. $H_0 : \mu_1 = \mu_2 \quad v.s. \quad H_a : \mu_1 \neq \mu_2,$
2. $H_0 : \mu_1 = \mu_3 \quad v.s. \quad H_a : \mu_1 \neq \mu_3,$
3. $H_0 : \mu_1 = \mu_4 \quad v.s. \quad H_a : \mu_1 \neq \mu_4,$
4. $H_0 : \mu_2 = \mu_3 \quad v.s. \quad H_a : \mu_2 \neq \mu_3,$
5. $H_0 : \mu_2 = \mu_4 \quad v.s. \quad H_a : \mu_2 \neq \mu_4,$
6. $H_0 : \mu_3 = \mu_4 \quad v.s. \quad H_a : \mu_3 \neq \mu_4.$

If we do each test at $\alpha = 0.05$, the Family-Wise Error Rate (FWER):

$$\text{FWER} = 1 - (1 - 0.05)^6 \approx 0.265$$

26.5% chance of at least one false positive!

The Alternative Procedure to t-Tests: Addressing the Multiple Comparisons Problem

Solution: Tukey's Honestly Significant Difference (HSD)

- Controls Family-Wise Error Rate (FWER) at α.
- Provides **simultaneous** confidence intervals.
- Valid for **balanced designs** (equal sample sizes).

Assumptions

1. **Normality**: $\varepsilon_{ij} \sim N(0, \sigma^2)$.
2. **Equal Variances**: $\sigma_1^2 = \sigma_2^2 = \cdots = \sigma_k^2$.
3. **Independence**: Observations are independent.

When to Use Tukey's HSD

- After significant one-way ANOVA.
- When comparing all possible pairs.
- When you want simultaneous confidence intervals.

Tukey's HSD Procedure for Multiple Comparison

Honestly Significant Difference (HSD)

$$\boxed{\text{HSD}_{ij} = q_\alpha(k, n-k) \times \sqrt{\frac{\text{MSE}}{2}\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}}$$

where:

- $q_\alpha(k, n-k)$: Critical value from Studentized Range Distribution with parameter values k and degrees of freedom $df = df_{SSE} = \nu = n - k$.
- k: Number of groups
- n_i and n_j: Sample sizes for groups i and j, respectively.
- $n - k$: Degrees of freedom for error (from ANOVA).
- MSE: Mean Square Error from ANOVA table.

Decision Rule for Pair (i, j)

Hypotheses:

$$H_0 : \mu_i = \mu_j \quad \text{vs} \quad H_a : \mu_i \neq \mu_j, \quad i, j = 1, \cdots, k, \quad i \neq j$$

Decision rule for pair (i, j)

Two group means $\bar{y}_i$ and $\bar{y}_j$ are significantly different if:

$$|\bar{y}_i - \bar{y}_j| > \text{HSD}_{ij}$$

Simultaneous Confidence Intervals

For any pair (i, j):

$$(\bar{y}_i - \bar{y}_j) \pm \text{HSD}_{ij}$$

These intervals have **simultaneous** coverage probability $1 - \alpha$.

Interpretation: We are 95% confident that in repeated sampling the value of $\mu_i - \mu_j$ would lie in the interval

$$(\bar{y}_i - \bar{y}_j) \pm q_\alpha(k, n-k) \cdot \sqrt{\frac{MSE}{2}\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}.$$

Tukey's HSD: Step-by-Step Procedure

1. Step 1: conduct ANOVA to test the null hypothesis $H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$.
   - Verify ANOVA is significant (reject H_0)
   - Obtain MSE from ANOVA table
   - Record degrees of freedom: k and $n - k$
2. Step 2: Calculate HSD

$$\text{HSD} = q_\alpha(k, n-k) \times \sqrt{\frac{\text{MSE}}{2}\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}$$

3. Step 3: Compute All Pairwise Differences:
   For all $i < j$:

$$d_{ij} = \bar{y}_i - \bar{y}_j$$

4. Step 4: Compare to HSD:
   For each pair (i, j):

$$|d_{ij}| > \text{HSD} \quad \Rightarrow \quad \text{Significant difference}$$

5. Step 5: Construct Confidence Intervals

$$d_{ij} \pm \text{HSD}$$

   If interval doesn't contain 0, difference is significant.

Complete Worked Example: Teaching Methods

Research Question: Do three different teaching methods affect exam scores?

- Method A: Traditional lecture.
- Method B: Online learning.
- Method C: Flipped classroom.

$n = 12$ students per method, $\alpha = 0.05$

Data Summary

	Method A	Method B	Method C
Sample size (n_i)	12	12	12
Sample mean ($\bar{y}_i$)	72.4	68.1	76.8
Sample variance (s_i^2)	64.0	49.0	81.0

ANOVA Results

Source	df	SS	MS	F
Treatments	2	568.9	284.45	5.88
Error	33	1596.0	48.36	
Total	35	2164.9		

$$F = 5.88, \quad p\text{-value} = 0.006, \quad \text{MSE} = 48.36$$

Example: Step 1 - Find Critical Value q

Parameters

- $k = 3$ groups
- $n = 12$ per group
- Total $n = 36$
- Error df: $n - k = 36 - 3 = 33$
- $\alpha = 0.05$

Finding $q_{0.05}(3, 33)$

- From Studentized Range Table:
- For $\nu = n - k = 30$: $q_{0.05}(3, 30) = 3.49$
- For $\nu = n - k = 40$: $q_{0.05}(3, 40) = 3.44$
- Interpolate for $\nu = n - k = 33$:

$$q_{0.05}(3, 40) \approx 3.49 - \frac{3}{10}(3.49 - 3.44) = 3.49 - 0.015 = 3.475$$

Step 2: Calculate HSD

$$\text{HSD} = 3.475 \times \sqrt{\frac{48.36}{2}(\frac{1}{12} + \frac{1}{12})} = 3.475 \times 2.008 = 6.98$$

Example: Step 3 - Compute Pairwise Differences

All Pairwise Differences

$$\begin{aligned} d_{AB} &= \bar{y}_A - \bar{y}_B = 72.4 - 68.1 = 4.3 \\ d_{AC} &= \bar{y}_A - \bar{y}_C = 72.4 - 76.8 = -4.4 \\ d_{BC} &= \bar{y}_B - \bar{y}_C = 68.1 - 76.8 = -8.7 \end{aligned}$$

Absolute values: $|d_{AB}| = 4.3$, $|d_{AC}| = 4.4$, $|d_{BC}| = 8.7$

Step 4: Compare to HSD

$$\text{HSD} = 6.98$$

- $H_0 : \mu_A = \mu_B$ vs $H_a : \mu_A \neq \mu_B \Rightarrow |d_{AB}| = 4.3 < 6.98$: Not significant
- $H_0 : \mu_A = \mu_C$ vs $H_a : \mu_A \neq \mu_C \Rightarrow |d_{AC}| = 4.4 < 6.98$: Not significant
- $H_0 : \mu_B = \mu_C$ vs $H_a : \mu_B \neq \mu_C \Rightarrow |d_{BC}| = 8.7 > 6.98$: Significant!

Interpretation

- Only Methods B and C differ significantly
- Method A is not significantly different from either B or C
- Method C (flipped classroom) has highest scores

Example: Step 5 - Simultaneous Confidence Intervals

95% simultaneous confidence intervals

For each difference:

$$d_{ij} \pm \text{HSD} = d_{ij} \pm 6.98$$

- $\mu_A - \mu_B$: $4.3 \pm 6.98 = (-2.68, 11.28)$ Contains 0
- $\mu_A - \mu_C$: $-4.4 \pm 6.98 = (-11.38, 2.58)$ Contains 0
- $\mu_B - \mu_C$: $-8.7 \pm 6.98 = (-15.68, -1.72)$ Does NOT contain 0

Interpretation of confidence intervals

- We are 95% confident that ALL THREE intervals contain their true differences.
- Only the B-C interval excludes 0.
- Method C scores are 1.72 to 15.68 points higher than Method B.

Kruskal-Wallis Test: Nonparametric Alternative to ANOVA

When to use

- Severe violation of normality (small n, skewed data).
- Severe violation of equal variances and when Outliers present which are strongly affect means.
- Ordinal data (ranks, Likert scales).

Hypotheses

H_0 : All k populations have the same median ($H_0 : Md_1 = \cdots = Md_k$)

H_a : At least two population medians are not the same ($H_0 : Md_i \neq Md_j$ for some $i \neq j$)

Key idea: rank-based analysis

- Replace actual values with ranks ($1 =$ smallest, $n =$ largest).
- If H_0 true: Ranks should be evenly distributed among groups.
- If H_a true: Some groups will have systematically higher/lower ranks.

Advantages

- Robust to outliers.
- No distributional assumptions.
- Works with ordinal data.

Kruskal-Wallis Test: Step-by-Step Procedure

1. **Step 1: Rank all observations and calculate rank sums**:
   - Combine all n observations, rank from 1 (smallest) to n (largest).
   - Calculate rank sums for each group:

$$T_{R_i} = \sum_{j=1}^{n_i} \text{rank}(y_{ij}) \quad \text{for each group } i = 1, \ldots, k$$

2. **Step 2: Compute test statistic**:

$$H = \frac{12}{n(n+1)} \sum_{i=1}^{k} \frac{T_{R_i}^2}{n_i} - 3(n+1)$$

   Note $H \sim \chi^2(k-1)$ for large samples (when $n_i \geq 5$) and $n = \sum_{i=1}^{k} n_i$.

3. **Step 3: Decision rule**: Reject H_0 if $H > \chi^2_{\alpha,k-1}$.

Denote average rank for ith sample by $\bar{R}_i = T_{R_i}/n_i$, the rank grand total by $T_R = \sum_{i=1}^{k} T_{R_i}$, and the overall average rank by $\bar{R} = \frac{1}{n} T_R$, so that

$$T_R = \sum_{i=1}^{n} T_{R_i} = \sum_{i=1}^{k} \sum_{j=1}^{n_i} R_{ij} = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

and

$$\bar{R} = \frac{1}{n} T_R = \frac{1}{n} \frac{n(n+1)}{2} = \frac{n+1}{2}$$

The Kruskal-Wallis test statistic can be written in different form as

$$\begin{aligned} H &= \frac{12}{n(n+1)} \sum_{i=1}^{k} \frac{T_{R_i}^2}{n_i} - 3(n+1) \\ &= \frac{12}{n(n+1)} \left\{ \sum_{i=1}^{k} n_i \bar{R}_i^2 - \frac{n(n+1)^2}{4} \right\} \\ &\overset{(\bar{R} = \frac{n+1}{2})}{\equiv} \frac{12}{n(n+1)} \left\{ \sum_{i=1}^{k} n_i \bar{R}_i^2 - n\bar{R}^2 \right\} \end{aligned}$$

Research Question

Do three different treatments (Drug A, Drug B, Placebo) differ in pain relief effectiveness?

Data: Pain Relief Scores (0-10, higher = better)

Drug A:	3	4	2	5	4	$n_1 = 5$
Drug B:	7	8	6	9		$n_2 = 4$
Placebo:	1	2	3	1	2	$n_3 = 5$

Parameters

- $k = 3$ groups (Drug A, Drug B, Placebo)
- $n_1 = 5, n_2 = 4, n_3 = 5$
- Total $n = 5 + 4 + 5 = 14$
- $\alpha = 0.05$

Step 1: Combine and Rank All Observations

Combine All Scores

All 14 scores: 3, 4, 2, 5, 4, 7, 8, 6, 9, 1, 2, 3, 1, 2

Sort and Assign Ranks (Handle Ties!)

Sorted Values	Frequency	Rank Range	Average Rank
1	2	$1-2$	$(1+2)/2 = 1.5$
2	3	$3-5$	$(3+4+5)/3 = 4$
3	2	$6-7$	$(6+7)/2 = 6.5$
4	2	$8-9$	$(8+9)/2 = 8.5$
5	1	10	10
6	1	11	11
7	1	12	12
8	1	13	13
9	1	14	14

Step 2: Assign Ranks Back to Original Groups

Drug A ($n = 5$)

Scores: $3, 4, 2, 5, 4$
Ranks: $6.5, 8.5, 4, 10, 8.5$
Sum of ranks: $R_1 = 6.5 + 8.5 + 4 + 10 + 8.5 = 37.5$

Drug B ($n = 4$)

Scores: $7, 8, 6, 9$
Ranks: $12, 13, 11, 14$
Sum of ranks: $R_2 = 12 + 13 + 11 + 14 = 50.0$

Placebo ($n = 5$)

Scores: $1, 2, 3, 1, 2$
Ranks: $1.5, 4, 6.5, 1.5, 4$
Sum of ranks: $R_3 = 1.5 + 4 + 6.5 + 1.5 + 4 = 17.5$

Verification

Total sum of ranks should be $\frac{n(n+1)}{2} = \frac{14 \times 15}{2} = 105$
$R_1 + R_2 + R_3 = 37.5 + 50.0 + 17.5 = 105$

Step 3: Calculate the Test Statistic H

Kruskal-Wallis Formula

$$H = \frac{12}{n(n+1)} \sum_{i=1}^{k} \frac{R_i^2}{n_i} - 3(N+1)$$

where:

- $n = 14$ (total observations)
- $k = 3$ (number of groups)
- R_i = sum of ranks for group i
- n_i = sample size for group i

Calculate Each Term

$$\frac{R_1^2}{n_1} = \frac{(37.5)^2}{5} = \frac{1406.25}{5} = 281.25$$
$$\frac{R_2^2}{n_2} = \frac{(50.0)^2}{4} = \frac{2500}{4} = 625.00$$
$$\frac{R_3^2}{n_3} = \frac{(17.5)^2}{5} = \frac{306.25}{5} = 61.25$$

Sum: $281.25 + 625.00 + 61.25 = 967.50$

Step 3 (cont.): Complete H Calculation

Plug into Formula

$$H = \frac{12}{14 \times 15} \times 967.50 - 3 \times 15$$

$$= \frac{12}{210} \times 967.50 - 45$$

$$= 0.057142857 \times 967.50 - 45$$

$$= 55.2857 - 45 = 10.2857 \approx 10.29$$

Step 4: Determine Critical Value and Make Decision

Chi-Square Distribution

Under H_0, $H \sim \chi^2_{k-1}$ for large samples
Degrees of freedom: $df = k - 1 = 3 - 1 = 2$

Critical Value

For $\alpha = 0.05$ and $df = 2$:

$$\chi^2_{0.05,2} = 5.991$$

Decision Rule

Reject H_0 if $H > \chi^2_{\alpha,k-1}$

$$10.29 > 5.991 \quad \Rightarrow \quad \text{Reject } H_0 \qquad (H_0 : Md_1 = Md_2 = Md_3)$$

Step 5: Interpret Results

Hypothesis Test Conclusion

We reject H_0 at $\alpha = 0.05$ level.
There is sufficient evidence to conclude that at least one treatment differs from the others in pain relief effectiveness.

Examine Rank Sums

- Drug B: Highest rank sum ($R_2 = 50.0$), average rank = 12.5
- Drug A: Moderate rank sum ($R_1 = 37.5$), average rank = 7.5
- Placebo: Lowest rank sum ($R_3 = 17.5$), average rank = 3.5

Dunn's Multiple Comparisons: Post-hoc for Kruskal-Wallis

When to use

After rejecting $H_0 : Md_1 = \cdots = Md_k$ in Kruskal-Wallis to identify which groups differ. It uses to compare average ranks $\bar{R}_i = R_i/n_i$ between groups.

Test Statistic for Pair i, j

Hypotheses:

$$H_0 : Md_i = Md_j, \quad \text{vs} \quad Md_i \neq Md_j, \quad i, j = 1, 2, \cdots, k, \quad i \neq j$$

$$z_{ij} = \frac{\bar{R}_i - \bar{R}_j}{\sqrt{\frac{n(n+1)}{12}\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}}$$

Simultaneous confidence intervals around $Md_i - Md_j$

$$(\bar{R}_i - \bar{R}_j) \pm z_{\alpha/(k(k-1))}\sqrt{\frac{n(n+1)}{12}\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}$$

Interpretation: We are 95% confident that in repeated sampling the value of $Md_i - Md_j$ would lie in the interval

$$(\bar{R}_i - \bar{R}_j) \pm z_{\alpha/(k(k-1))}\sqrt{\frac{n(n+1)}{12}\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}$$

From Previous Example

Average Ranks

$$\bar{R}_1 = \frac{37.5}{5} = 7.5, \quad \bar{R}_2 = \frac{50.0}{4} = 12.5, \quad \bar{R}_3 = \frac{17.5}{5} = 3.5$$

$$k = 3, \alpha = 0.05, \quad \frac{\alpha}{k(k-1)} = \frac{0.05}{6} = 0.00833$$

$$z_{0.00833} \approx 2.39 \text{ (from normal table)}$$

$$\sqrt{\frac{n(n+1)}{12}} = \sqrt{\frac{14 \times 15}{12}} = \sqrt{17.5} = 4.1833$$

Dunn's Test Calculations

Pairwise Comparisons

1. **Drug A vs Drug B**: $H_0 : Md_1 = Md_2$ vs $H_a : Md_1 \neq Md_2$

$$z_{12} = \frac{7.5 - 12.5}{4.1833 \times \sqrt{\frac{1}{5} + \frac{1}{4}}} = \frac{-5.0}{4.1833 \times \sqrt{0.45}} = \frac{-5.0}{4.1833 \times 0.6708} = \frac{-5.0}{2.806} = -1.78$$

$|z_{12}| = 1.78 < 2.39$: Not significant (fail to reject H_0)

2. **Drug A vs Placebo**: $H_0 : Md_1 = Md_3$ vs $H_a : Md_1 \neq Md_3$

$$z_{13} = \frac{7.5 - 3.5}{4.1833 \times \sqrt{\frac{1}{5} + \frac{1}{5}}} = \frac{4.0}{4.1833 \times \sqrt{0.4}} = \frac{4.0}{4.1833 \times 0.6325} = \frac{4.0}{2.646} = 1.51$$

$|z_{13}| = 1.51 < 2.39$: Not significant (fail to reject H_0)

3. **Drug B vs Placebo**: $H_0 : Md_2 = Md_3$ vs $H_a : Md_2 \neq Md_3$

$$z_{23} = \frac{12.5 - 3.5}{4.1833 \times \sqrt{\frac{1}{4} + \frac{1}{5}}} = \frac{9.0}{4.1833 \times \sqrt{0.45}} = \frac{9.0}{4.1833 \times 0.6708} = \frac{9.0}{2.806} = 3.21$$

$|z_{23}| = 3.21 > 2.39$: Significant! (reject H_0)

Final Conclusions and Practical Implications

Dunn's Test Results Summary

- Drug A vs Drug B: Not significantly different ($z = -1.78$, $p > 0.05$)
- Drug A vs Placebo: Not significantly different ($z = 1.51$, $p > 0.05$)
- Drug B vs Placebo: Significantly different ($z = 3.21$, $p < 0.05$)

Practical Interpretation

- Drug B shows significantly better pain relief than placebo
- Drug A is not significantly better than placebo
- No significant difference between Drug A and Drug B
- Possible reasons: Drug A ineffective, small sample size, measurement issues

Example

The achievement test scores for four different groups of students, each group taught by a different teaching technique, are collected using a complete randomized design. These data are shown in the table below.

Teaching Techniques			
1	2	3	4
65	75	59	94
87	69	78	89
73	83	67	80
79	81	62	88
81	72	83	
69	79	76	
	90		

(a) Test whether the average math scores of students are the same for the four different techniques. List all necessary assumptions and check any suspect assumption.

(b) If the average math scores for the **four** groups are found to be significantly different in (a), do a follow-up analysis to determine which are the same and which differ.

(c) Perform a non-parametric analysis for the tasks (a) and (b).

Use $\alpha = 0.10$ in all the tests.

Solution (a) - ANOVA Assumptions

The data are collected using CRD.

- The hypothesis testing problem is

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

$$\text{v.s.} \quad H_a : \text{not all } \mu_i\text{'s are the same.}$$

  We use ANOVA F test.

- The assumptions for the F test:
  (i) The four samples are independent random samples from the four treatment populations (CRD).
  (ii) All treatment populations have normal distributions.
  (iii) All treatment populations have the same variance.

Solution (a) - Checking Assumptions

Assumption (i) is ensured by CRD. Assumption (ii) can be checked using histograms (Not required here).
We now check the constant variance assumption using **Hartley's test**.

- Testing problem:

$$H_0 : \sigma_1^2 = \sigma_2^2 = \sigma_3^2 = \sigma_4^2 \quad \text{v.s.} \quad H_a : \text{not all } \sigma_i^2 \text{ are the same}$$

- Finding test statistic:

$$n_1 = 6, \quad n_2 = 7, \quad n_3 = 6, \quad n_4 = 4$$

$$\Rightarrow \overline{n} = \frac{1}{k}\sum_{i=1}^{k} n_i = 5.75, \quad [\overline{n}] = 5$$

$$S_i^2 = \frac{1}{n_i - 1}\left\{\sum_{j=1}^{n_i} y_{ij}^2 - \frac{1}{n_i}\left(\sum_{j=1}^{n_i} y_{ij}\right)^2\right\} \approx 66.667$$

Solution (a) - Hartley's Test (Continued)

$$S_2^2 = \frac{1}{n_2 - 1}\left\{\sum_{j=1}^{n_2} y_{2j}^2 - \frac{1}{n_2}\left(\sum_{j=1}^{n_2} y_{2j}\right)^2\right\} \approx 50.619,$$

$$S_3^2 = \text{ Formula here } \approx 91.767, \quad S_4^2 = \text{ Formula here } \approx 33.583.$$

Hence $S_{\max}^2 = S_3^2 \approx 91.767$ and $S_{\min}^2 = S_4^2 \approx 33.583$.
Test statistic:

$$F_{\max} = \frac{S_{\max}^2}{S_{\min}^2} \approx \frac{91.767}{33.583} \approx 2.733.$$

- Using the Hartley's table for $\alpha = 0.05$
- $F_{\max} < F_{\max}(k, [\overline{n}] - 1; \alpha) = F_{\max}(4, 4; 0.05) = 20.6$ (approx. from Hartley's table)

Since $F_{\max} \approx 2.733 < 20.6$, we do not reject H_0 at the 5% level and conclude that σ_i^2's are the same.

Solution (a) - ANOVA F-Test

Since we do not find violation of constant variance assumption, we can proceed with the F-test.

- ANOVA table:

$$n = \sum_{i=1}^{k} n_i = 23, \quad T_1 = \sum_{j=1}^{n_1} y_{1j} = 454,$$

$$T_2 = \sum_{j=1}^{n_2} y_{2j} = 549, \quad T_3 = \sum_{j=1}^{n_3} y_{3j} = 425, \quad T_4 = \sum_{j=1}^{n_4} y_{4j} = 351.$$

$$TSS = \sum_{i=1}^{k} \sum_{j=1}^{n_i} y_{ij}^2 - \frac{(\sum_{i=1}^{k} \sum_{j=1}^{n_i} y_{ij})^2}{n} \approx 1909.2174.$$

$$df_{TSS} = n - 1 = 22$$

$$SS_{Tr} = \sum_{i=1}^{k} \frac{T_i^2}{n_i} - \frac{(\sum_{i=1}^{k} T_i)^2}{n} \approx 712.5864$$

$$df_{SS_{Tr}} = k - 1 = 3; \quad MS_{Tr} = \frac{SS_{Tr}}{df_{SS_{Tr}}} \approx 237.529$$

$$SSE = TSS - SS_{Tr} \approx 1196.6310$$

$$df_{SSE} = n - k = 20; \quad MSE = \frac{SSE}{df_{SSE}} \approx 59.8316$$

Solution (a) - ANOVA F-Test (Continued)

The ANOVA table is

Source	d.f.	SS	MS	F
Treatments	3	712.5864	237.529	3.97
Error	20	1196.6310	59.832	
Total	22	1909.2174		

- The hypothesis testing problem:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 \quad \text{vs.} \quad H_a : \text{not all } \mu_i\text{'s are the same}$$

- Test statistic for F test:

$$F_T = \frac{MSTr}{MSE} \approx 3.97.$$

Under H_0, $F_T \sim F_{k-1,\, n-k} = F_{3,\, 20}$.

- We reject H_0 at the $\alpha = 0.10$ significance level if $F_T > F_{k-1,\, n-k;\, \alpha} = F_{3,\, 20;\, 0.10} = 2.38$.
- Since $F_T \approx 3.97 > F_{3,\, 20;\, 0.10} = 2.38$, we reject H_0 in favor of H_a at the 0.10 significance level and conclude that the average mean math scores under the four teaching techniques differ.

Solution (b) - Tukey's HSD Follow-up Analysis

Since we found differences in average math scores for the four teaching techniques, we use **Tukey's HSD** (Honestly Significant Difference) to find out which pairs of average scores are different.

- There are
$$\binom{k}{2} = \binom{4}{2} = \frac{4 \times 3}{2} = 6 \text{ pairs}$$
of μ_i and μ_j to compare.
- The hypothesis testing problems are
$$H_0 : \mu_i = \mu_j \quad \text{v.s.} \quad H_a : \mu_i \neq \mu_j$$
for $i, j = 1, 2, 3, 4$, $i \neq j$.

Solution (b) - Tukey's HSD Critical Values

The critical value for a given pair of i and j is:

$$\text{HSD}_{ij} = q_\alpha(k, n-k)\sqrt{\frac{MSE}{2}\left(\frac{1}{n_i} + \frac{1}{n_j}\right)} \approx (3.24)\sqrt{29.916\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}$$

$$\text{HSD}_{12} = (3.24)\sqrt{29.916\left(\frac{1}{6} + \frac{1}{7}\right)} \approx 10.07$$

$$\text{HSD}_{13} = (3.24)\sqrt{29.916\left(\frac{1}{6} + \frac{1}{6}\right)} \approx 10.23$$

$$\text{HSD}_{14} = (3.24)\sqrt{29.916\left(\frac{1}{6} + \frac{1}{4}\right)} \approx 11.45$$

$$\text{HSD}_{23} = (3.24)\sqrt{29.916\left(\frac{1}{7} + \frac{1}{6}\right)} \approx 10.07$$

$$\text{HSD}_{24} = (3.24)\sqrt{29.916\left(\frac{1}{7} + \frac{1}{4}\right)} \approx 11.18$$

$$\text{HSD}_{34} = (3.24)\sqrt{29.916\left(\frac{1}{6} + \frac{1}{4}\right)} \approx 11.45$$

Solution (b) - Pairwise Comparisons

$$\bar{y}_1 = \frac{T_1}{n_1} = \frac{454}{6} \approx 75.667, \quad \bar{y}_2 = \frac{T_2}{n_2} = \frac{549}{7} \approx 78.429$$

$$\bar{y}_3 = \frac{T_3}{n_3} = \frac{425}{6} \approx 70.833, \quad \bar{y}_4 = \frac{T_4}{n_4} = \frac{351}{4} = 87.75$$

$$|\bar{y}_1 - \bar{y}_2| = 2.762 < \text{HSD}_{12} \approx 10.07 \quad \text{(do not reject } H_0\text{)}$$
$$|\bar{y}_1 - \bar{y}_3| = 4.834 < \text{HSD}_{13} \approx 10.23 \quad \text{(do not reject } H_0\text{)}$$
$$|\bar{y}_1 - \bar{y}_4| = 12.083 > \text{HSD}_{14} \approx 11.45 \quad \text{(reject } H_0\text{)}$$
$$|\bar{y}_2 - \bar{y}_3| = 7.596 < \text{HSD}_{23} \approx 10.07 \quad \text{(do not reject } H_0\text{)}$$
$$|\bar{y}_2 - \bar{y}_4| = 9.321 < \text{HSD}_{24} \approx 11.18 \quad \text{(do not reject } H_0\text{)}$$
$$|\bar{y}_3 - \bar{y}_4| = 16.917 > \text{HSD}_{34} \approx 11.45 \quad \text{(reject } H_0\text{)}$$

Hence, we conclude that $\mu_1 \neq \mu_4$ and $\mu_3 \neq \mu_4$, i.e., there are differences between average math scores for:

- Teaching technique 1 and technique 4
- Teaching technique 3 and technique 4

Notes on Multiple Comparisons

(1) The pairwise comparisons results using **Fisher's LSD method** and **Bonferroni's method** are different from the result using **Tukey's HSD**. We will use Tukey's HSD unless otherwise specified.

(2) We perform pairwise comparisons only when $H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$ in the F test is **rejected**.

If H_0 in the F test is not rejected, multiple comparisons of pairwise differences are not needed.

Solution (c) - Non-parametric Analysis (Kruskal-Wallis Test)

The ranks of the data in the pooled sample are given in parentheses "()" in the table below.

Teaching Techniques			
1	**2**	**3**	**4**
65 (3)	75 (9)	59 (1)	94 (23)
87 (19)	69 (5.5)	78 (11)	89 (21)
73 (8)	83 (17.5)	67 (4)	80 (14)
79 (12.5)	81 (15.5)	62 (2)	88 (20)
81 (15.5)	72 (7)	83 (17.5)	
69 (5.5)	79 (12.5)	76 (10)	
	90 (22)		

$$T_{R_1} = \sum_{j=1}^{n_1} R_{1j} = 63.5, \quad T_{R_2} = \sum_{j=1}^{n_2} R_{2j} = 89,$$

$$T_{R_3} = \sum_{j=1}^{n_3} R_{3j} = 45.5, \quad T_{R_4} = \sum_{j=1}^{n_4} R_{4j} = 78.$$

Solution (c) - Kruskal-Wallis Test Results

1. **Hypothesis testing problem:**

$$H_0 : Md_1 = Md_2 = Md_3 = Md_4$$

$$\text{v.s.} \quad H_a : \text{not all } Md_i\text{'s are the same}$$

2. **Test statistic for Kruskal-Wallis test:**

$$H = \frac{12}{n(n+1)} \sum_{i=1}^{4} \frac{T_{R_i}^2}{n_i} - 3(n+1)$$

$$H = \frac{12}{23(24)} \left(\frac{(63.5)^2}{6} + \frac{(89)^2}{7} + \frac{(45.5)^2}{6} + \frac{(78)^2}{4} \right) - 3(24)$$

$$H \approx 7.775$$

Under H_0, H is approximately distributed as $\chi^2_{k-1} = \chi^2_3$.

3. **Decision rule:** We reject H_0 if $H > \chi^2_{k-1,\,\alpha} = \chi^2_{3;\,0.10} \approx 6.251$.
4. **Conclusion:** Since $H \approx 7.775 > \chi^2_{3;\,0.10} = 6.251$, we reject H_0 in favor of H_a at the 0.10 significance level and conclude that there are differences among the four teaching techniques.

Solution (c) - Follow-up Analysis

Since H_0 in Kruskal-Wallis is rejected, we perform a follow-up analysis using **Dunn's procedure** to find out pairwise differences.

This is left as an exercise.

Important Note

If $H_0 : Md_1 = Md_2 = \cdots = Md_4$ is **not rejected** in the Kruskal-Wallis test, then a follow-up analysis comparing pairwise medians is **not necessary**.

Comprehensive Comparison: When to Use What

One-Way ANOVA (Parametric)	Kruskal-Wallis (Nonparametric)
Assumptions: 1. Normality 2. Equal variances 3. Independent samples	**Assumptions:** 1. Independent samples 2. Continuous or ordinal data 3. Same shape distributions (for medians interpretation)
Hypotheses: $H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$ H_a: At least one μ_i different	**Hypotheses:** H_0: All populations have same distribution H_a: At least one distribution shifted
Strengths: - More power when assumptions met - Provides estimates of means - Well-understood	**Strengths:** - Robust to outliers - No normality assumption - Works with ordinal data
Weaknesses: - Sensitive to violations - Requires interval data	**Weaknesses:** - Less power when ANOVA assumptions hold - Doesn't estimate means directly
Multiple Comparisons: Tukey's HSD	**Multiple Comparisons:** Dunn's test

Common Pitfalls

Common mistakes to avoid

1. Using multiple t-tests without adjustment (controls per-comparison error rate, not family-wise)
2. Interpreting significant ANOVA as "all groups differ" (need post-hoc tests)
3. Using Tukey's HSD with unbalanced designs (use Bonferroni instead)
4. Ignoring assumption checks (always check normality and equal variances)
5. Misinterpreting Kruskal-Wallis (tests distributions, not just medians)

Exam strategy

- **Step 1:** State hypotheses clearly (H_0, H_a).
- **Step 2:** Check assumptions explicitly (Kruskal-Wallis is not needed).
- **Step 3:** Choose appropriate test (justify choice).
- **Step 4:** Compute test statistic (show formulas).
- **Step 5:** Make decision (reject/fail to reject).
- **Step 6:** Interpret in context (plain English).

Chi-Square Table

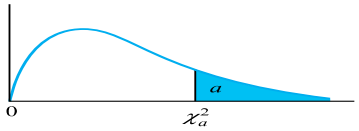

TABLE 5
Critical Values of Chi-Square

df	$\chi^2_{.995}$	$\chi^2_{.990}$	$\chi^2_{.975}$	$\chi^2_{.950}$	$\chi^2_{.900}$
1	.0000393	.0001571	.0009821	.0039321	.0157908
2	.0100251	.0201007	.0506356	.102587	.210720
3	.0717212	.114832	.215795	.351846	.584375
4	.206990	.297110	.484419	.710721	1.063623
5	.411740	.554300	.831211	1.145476	1.61031
6	.675727	.872085	1.237347	1.63539	2.20413
7	.989265	1.239043	1.68987	2.16735	2.83311
8	1.344419	1.646482	2.17973	2.73264	3.48954
9	1.734926	2.087912	2.70039	3.32511	4.16816
10	2.15585	2.55821	3.24697	3.94030	4.86518
11	2.60321	3.05347	3.81575	4.57481	5.57779
12	3.07382	3.57056	4.40379	5.22603	6.30380
13	3.56503	4.10691	5.00874	5.89186	7.04150
14	4.07468	4.66043	5.62872	6.57063	7.78953
15	4.60094	5.22935	6.26214	7.26094	8.54675
16	5.14224	5.81221	6.90766	7.96164	9.31223
17	5.69724	6.40776	7.56418	8.67176	10.0852
18	6.26481	7.01491	8.23075	9.39046	10.8649
19	6.84398	7.63273	8.90655	10.1170	11.6509
20	7.43386	8.26040	9.59083	10.8508	12.4426
21	8.03366	8.89720	10.28293	11.5913	13.2396
22	8.64272	9.54249	10.9823	12.3380	14.0415
23	9.26042	10.19567	11.6885	13.0905	14.8479
24	9.88623	10.8564	12.4011	13.8484	15.6587
25	10.5197	11.5240	13.1197	14.6114	16.4734
26	11.1603	12.1981	13.8439	15.3791	17.2919
27	11.8076	12.8786	14.5733	16.1513	18.1138
28	12.4613	13.5648	15.3079	16.9279	18.9392
29	13.1211	14.2565	16.0471	17.7083	19.7677
30	13.7867	14.9535	16.7908	18.4926	20.5992
40	20.7065	22.1643	24.4331	26.5093	29.0505
50	27.9907	29.7067	32.3574	34.7642	37.6886
60	35.5346	37.4848	40.4817	43.1879	46.4589
70	43.2752	45.4418	48.7576	51.7393	55.3290
80	51.1720	53.5400	57.1532	60.3915	64.2778
90	59.1963	61.7541	65.6466	69.1260	73.2912
100	67.3276	70.0648	74.2219	77.9295	82.3581

Source: From "Tables of the Percentage Points of the χ^2-Distribution," *Biometrika Tables for Statisticians,* Vol. 1, 3rd ed. (1966). Reproduced by permission of the *Biometrika* Trustees.

TABLE 5 *(continued)*

$\chi^2_{.100}$	$\chi^2_{.050}$	$\chi^2_{.025}$	$\chi^2_{.010}$	$\chi^2_{.005}$	*df*
2.70554	3.84146	5.02389	6.63490	7.87944	1
4.60517	5.99147	7.37776	9.21034	10.5966	2
6.25139	7.81473	9.34840	11.3449	12.8381	3
7.77944	9.48773	11.1433	13.2767	14.8602	4
9.23635	11.0705	12.8325	15.0863	16.7496	5
10.6446	12.5916	14.4494	16.8119	18.5476	6
12.0170	14.0671	16.0128	18.4753	20.2777	7
13.3616	15.5073	17.5346	20.0902	21.9550	8
14.6837	16.9190	19.0228	21.6660	23.5893	9
15.9871	18.3070	20.4831	23.2093	25.1882	10
17.2750	19.6751	21.9200	24.7250	26.7569	11
18.5494	21.0261	23.3367	26.2170	28.2995	12
19.8119	22.3621	24.7356	27.6883	29.8194	13
21.0642	23.6848	26.1190	29.1413	31.3193	14
22.3072	24.9958	27.4884	30.5779	32.8013	15
23.5418	26.2962	28.8485	31.9999	34.2672	16
24.7690	27.8571	30.1910	33.4087	35.7185	17
25.9894	28.8693	31.5264	34.8053	37.1564	18
27.2036	30.1435	32.8523	36.1908	38.5822	19
28.4120	31.4104	34.1696	37.5662	39.9968	20
29.6151	32.6705	35.4789	38.9321	41.4010	21
30.8133	33.9244	36.7807	40.2894	42.7956	22
32.0069	35.1725	38.0757	41.6384	44.1813	23
33.1963	36.4151	39.3641	42.9798	45.5585	24
34.3816	37.6525	40.6465	44.3141	46.9278	25
35.5631	38.8852	41.9232	45.6417	48.2899	26
36.7412	40.1133	43.1944	46.9630	49.6449	27
37.9159	41.3372	44.4607	48.2782	50.9933	28
39.0875	42.5569	45.7222	49.5879	52.3356	29
40.2560	43.7729	46.9792	50.8922	53.6720	30
51.8050	55.7585	59.3417	63.6907	66.7659	40
63.1671	67.5048	71.4202	76.1539	79.4900	50
74.3970	79.0819	83.2976	88.3794	91.9517	60
85.5271	90.5312	95.0231	100.425	104.215	70
96.5782	101.879	106.629	112.329	116.321	80
107.565	113.145	118.136	124.116	128.299	90
118.498	124.342	129.561	135.807	140.169	100

Hartley's Table

TABLE 14 Percentage Points of $F_{max} = s^2_{max}/s^2_{min}$ (Hartley's)

Upper 5% points

df_2 \ t	2	3	4	5	6	7	8	9	10	11	12
2	39.0	87.5	142	202	266	333	403	475	550	626	704
3	15.4	27.8	39.2	50.7	62.0	72.9	83.5	93.9	104	114	124
4	9.60	15.5	20.6	25.2	29.5	33.6	37.5	41.1	44.6	48.0	51.4
5	7.15	10.8	13.7	16.3	18.7	20.8	22.9	24.7	26.5	28.2	29.9
6	5.82	8.38	10.4	12.1	13.7	15.0	16.3	17.5	18.6	19.7	20.7
7	4.99	6.94	8.44	9.70	10.8	11.8	12.7	13.5	14.3	15.1	15.8
8	4.43	6.00	7.18	8.12	9.03	9.78	10.5	11.1	11.7	12.2	12.7
9	4.03	5.34	6.31	7.11	7.80	8.41	8.95	9.45	9.91	10.3	10.7
10	3.72	4.85	5.67	6.34	6.92	7.42	7.87	8.28	8.66	9.01	9.34
12	3.28	4.16	4.79	5.30	5.72	6.09	6.42	6.72	7.00	7.25	7.48
15	2.86	3.54	4.01	4.37	4.68	4.95	5.19	5.40	5.59	5.77	5.93
20	2.46	2.95	3.29	3.54	3.76	3.94	4.10	4.24	4.37	4.49	4.59
30	2.07	2.40	2.61	2.78	2.91	3.02	3.12	3.21	3.29	3.36	3.39
60	1.67	1.85	1.96	2.04	2.11	2.17	2.22	2.26	2.30	2.33	2.36
∞	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Upper 1% points

df_2 \ t	2	3	4	5	6	7	8	9	10	11	12
2	199	448	729	1036	1362	1705	2063	2432	2813	3204	3605
3	47.5	85	120	151	184	21(6)	24(9)	28(1)	31(0)	33(7)	36(1)
4	23.2	37	49	59	69	79	89	97	106	113	120
5	14.9	22	28	33	38	42	46	50	54	57	60
6	11.1	15.5	19.1	22	25	27	30	32	34	36	37
7	8.89	12.1	14.5	16.5	18.4	20	22	23	24	26	27
8	7.50	9.9	11.7	13.2	14.5	15.8	16.6	17.9	18.9	19.8	21
9	6.54	8.5	9.9	11.1	12.1	13.1	13.9	14.7	15.3	16.0	16.6
10	5.85	7.4	8.6	9.6	10.4	11.1	11.8	12.4	12.9	13.4	13.9
12	4.91	6.1	6.9	7.6	8.2	8.7	9.1	9.5	9.9	10.2	10.6
15	4.07	4.9	5.5	6.0	6.4	6.7	7.1	7.3	7.5	7.8	8.0
20	3.32	3.8	4.3	4.6	4.9	5.1	5.3	5.5	5.6	5.8	5.9
30	2.63	3.0	3.3	3.4	3.6	3.7	3.8	3.9	4.0	4.1	4.2
60	1.96	2.2	2.3	2.4	2.4	2.5	2.5	2.6	2.6	2.7	2.7
∞	1.00	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0

s^2_{max} is the largest and s^2_{min} the smallest in a set of t independent mean squares, each based on $df_2 = n - 1$ degrees of freedom. Values in the column $t = 2$ and in the rows $df_2 = 2$ and ∞ are exact. Elsewhere the third digit may be in error by a few units for the 5% points and several units for the 1% points. The third digit figures in parentheses for $df_2 = 3$ are the most uncertain. From *Biometrika Tables for Statisticians*, 3rd ed., Vol. 1, edited by E. S. Pearson and H. O. Hartley (New York: Cambridge University Press, 1966), Table, p. 202. Reproduced by permission of the *Biometrika Trustees*.

Tukey Table

Tukey Test Critical Values Table

1

α = 0.01																				
		k																		
		2	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*	*13*	*14*	*15*	*16*	*17*	*18*	*19*	*20*
v	*1*	90.03	135.0	164.3	185.6	202.2	215.8	227.2	237.0	245.6	253.2	260.0	266.2	271.8	277.0	281.8	286.3	290.4	294.3	298.0
	2	14.04	19.02	22.29	24.72	26.63	28.20	29.53	30.68	31.69	32.59	33.40	34.13	34.81	35.43	36.00	36.53	37.03	37.50	37.95
	3	8.26	10.62	12.17	13.33	14.24	15.00	15.64	16.20	16.69	17.13	17.53	17.89	18.22	18.52	18.81	19.07	19.32	19.55	19.77
	4	6.51	8.12	9.17	9.96	10.58	11.10	11.55	11.93	12.27	12.57	12.84	13.09	13.32	13.53	13.73	13.91	14.08	14.24	14.40
	5	5.70	6.98	7.80	8.42	8.91	9.32	9.67	9.97	10.24	10.48	10.70	10.89	11.08	11.24	11.40	11.55	11.68	11.81	11.93
	6	5.24	6.33	7.03	7.56	7.97	8.32	8.61	8.87	9.10	9.30	9.48	9.65	9.81	9.95	10.08	10.21	10.32	10.43	10.54
	7	4.95	5.92	6.54	7.01	7.37	7.68	7.94	8.17	8.37	8.55	8.71	8.86	9.00	9.12	9.24	9.35	9.46	9.55	9.65
	8	4.75	5.64	6.20	6.62	6.96	7.24	7.47	7.68	7.86	8.03	8.18.	8.31	8.44	8.55	8.66	8.76	8.85	8.94	9.03
	9	4.60	5.43	5.96	6.35	6.66	6.91	7.13	7.33	7.49	7.65	7.78	7.91	8.03	8.13	8.23	8.33	8.41	8.49	8.57
	10	4.48	5.27	5.77	6.14	6.43	6.67	6.87	7.05	7.21	7.36	7.49	7.60	7.71	7.81	7.91	7.99	8.08	8.15	8.23
	11	4.39	5.15	5.62	5.97	6.25	6.48	6.67	6.84	6.99	7.13	7.25	7.36	7.46	7.56	7.65	7.73	7.81	7.88	7.95
	12	4.32	5.05	5.50	5.84	6.10	6.32	6.51	6.67	6.81	6.94	7.06	7.17	7.26	7.36	7.44	7.52	7.59	7.66	7.73
	13	4.26	4.96	5.40	5.73	5.98	6.19	6.37	6.53	6.67	6.79	6.90	7.01	7.10	7.19	7.27	7.35	7.42	7.48	7.55
	14	4.21	4.89	5.32	5.63	5.88	6.08	6.26	6.41	6.54	6.66	6.77	6.87	6.96	7.05	7.13	7.20	7.27	7.33	7.39
	15	4.17	4.84	5.25	5.56	5.80	5.99	6.16	6.31	6.44	6.55	6.66	6.76	6.84	6.93	7.00	7.07	7.14	7.20	7.26
	16	4.13	4.79	5.19	5.49	5.72	5.92	6.08	6.22	6.35	6.46	6.56	6.66	6.74	6.82	6.90	6.97	7.03	7.09	7.15
	17	4.10	4.74	5.14	5.43	5.66	5.85	6.01	6.15	6.27	6.38	6.48	6.57	6.66	6.73	6.81	6.87	6.94	7.00	7.05
	18	4.07	4.70	5.09	5.38	5.60	5.79	5.94	6.08	6.20	6.31	6.41	6.50	6.58	6.65	6.73	6.79	6.85	6.91	6.97
	19	4.05	4.67	5.05	5.33	5.55	5.73	5.89	6.02	6.14	6.25	6.34	6.43	6.51	6.58	6.65	6.72	6.78	6.84	6.89
	20	4.02	4.64	5.02	5.29	5.51	5.69	5.84	5.97	6.09	6.19	6.28	6.37	6.45	6.52	6.59	6.65	6.71	6.77	6.82
	24	3.96	4.55	4.91	5.17	5.37	5.54	5.69	5.81	5.92	6.02	6.11	6.19	6.26	6.33	6.39	6.45	6.51	6.56	6.61
	30	3.89	4.45	4.80	5.05	5.24	5.40	5.54	5.65	5.76	5.85	5.93	6.01	6.08	6.14	6.20	6.26	6.31	6.36	6.41
	40	3.82	4.37	4.70	4.93	5.11	5.26	5.39	5.50	5.60	5.69	5.76	5.83	5.90	5.96	6.02	6.07	6.12	6.16	6.21
	60	3.76	4.28	4.59	4.82	4.99	5.13	5.25.	5.36	5.45	5.53	5.60	5.67	5.73	5.78	5.84	5.89	5.93	5.97	6.01
	120	3.70	4.20	4.50	4.71	4.87	5.01	5.12	5.21	5.30	5.37	5.44	5.50	5.56	5.61	5.66	5.71	5.75	5.79	5.83
	∞	3.64	4.12	4.40	4.60	4.76	4.88	4.99	5.08	5.16	5.23	5.29	5.35	5.40	5.45	5.49	5.54	5.57	5.61	5.65

Tukey Test Critical Values Table

$\alpha = 0.05$																				
		k																		
		2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
v	1	17.97	26.98	32.82	37.08	40.41	43.12	45.40	47.36	49.07	50.59	51.96	53.20	54.33	55.36	56.32	57.22	58.04	58.83	59.56
	2	6.08	8.33	9.80	10.88	11.74	12.44	13.03	13.54	13.99	14.39	14.75	15.08	15.38	15.65	15.91	16.14	16.37	16.57	16.77
	3	4.50	5.91	6.82	7.50	8.04	8.48	8.85	9.18	9.46	9.72	9.95	10.15	10.35	10.53	10.69	10.84	10.98	11.11	11.24
	4	3.93	5.04	5.76	6.29	6.71	7.05	7.35	7.60	7.83	8.03	8.21	8.37	8.52	8.66	8.79	8.91	9.03	9.13	9.23
	5	3.64	4.60	5.22	5.67	6.03	6.33	6.58	6.80	6.99	7.17	7.32	7.47	7.60	7.72	7.83	7.93	8.03	8.12	8.21
	6	3.46	4.34	4.90	5.30	5.63	5.90	6.12	6.32	6.49	6.65	6.79	6.92	7.03	7.14	7.24	7.34	7.43	7.51	7.59
	7	3.34	4.16	4.68	5.06	5.36	5.61	5.82	6.00	6.16	6.30	6.43	6.55	6.66	6.76	6.85	6.94	7.02	7.10	7.17
	8	3.26	4.04	4.53	4.89	5.17	5.40	5.60	5.77	5.92	6.05	6.18	6.29	6.39	6.48	6.57	6.65	6.73	6.80	6.87
	9	3.20	3.95	4.41	4.76	5.02	5.24	5.43	5.59	5.74	5.87	5.98	6.09	6.19	6.28	6.36	6.44	6.51	6.58	6.64
	10	3.15	3.88	4.33	4.65	4.91	5.12	5.30	5.46	5.60	5.72	5.83	5.93	6.03	6.11	6.19	6.27	6.34	6.40	6.47
	11	3.11	3.82	4.26	4.57	4.82	5.03	5.20	5.35	5.49	5.61	5.71	5.81	5.90	5.98	6.06	6.13	6.20	6.27	6.33
	12	3.08	3.77	4.20	4.51	4.75	4.95	5.12	5.27	5.39	5.51	5.61	5.71	5.80	5.88	5.95	6.02	6.09	6.15	6.21
	13	3.06	3.73	4.15	4.45	4.69	4.88	5.05	5.19	5.32	5.43	5.53	5.63	5.71	5.79	5.86	5.93	5.99	6.05	6.11
	14	3.03	3.70	4.11	4.41	4.64	4.83	4.99	5.13	5.25	5.36	5.46	5.55	5.64	5.71	5.79	5.85	5.91	5.97	6.03
	15	3.01	3.67	4.08	4.37	4.59	4.78	4.94	5.08	5.20	5.31	5.40	5.49	5.57	5.65	5.72	5.78	5.85	5.90	5.96
	16	3.00	3.65	4.05	4.33	4.56	4.74	4.90	5.03	5.15	5.26	5.35	5.44	5.52	5.59	5.66	5.73	5.79	5.84	5.90
	17	2.98	3.63	4.02	4.30	4.52	4.70	4.86	4.99	5.11	5.21	5.31	5.39	5.47	5.54	5.61	5.67	5.73	5.79	5.84
	18	2.97	3.61	4.00	4.28	4.49	4.67	4.82	4.96	5.07	5.17	5.27	5.35	5.43	5.50	5.57	5.63	5.69	5.74	5.79
	19	2.96	3.59	3.98	4.25	4.47	4.65	4.79	4.92	5.04	5.14	5.23	5.31	5.39	5.46	5.53	5.59	5.65	5.70	5.75
	20	2.95	3.58	3.96	4.23	4.45	4.62	4.77	4.90	5.01	5.11	5.20	5.28	5.36	5.43	5.49	5.55	5.61	5.66	5.71
	24	2.92	3.53	3.90	4.17	4.37	4.54	4.68	4.81	4.92	5.01	5.10	5.18	5.25	5.32	5.38	5.44	5.49	5.55	5.59
	30	2.89	3.49	3.85	4.10	4.30	4.46	4.60	4.72	4.82	4.92	5.00	5.08	5.15	5.21	5.27	5.33	5.38	5.43	5.47
	40	2.86	3.44	3.79	4.04	4.23	4.39	4.52	4.63	4.73	4.82	4.90	4.98	5.04	5.11	5.16	5.22	5.27	5.31	5.36
	60	2.83	3.40	3.74	3.98	4.16	4.31	4.44	4.55	4.65	4.73	4.81	4.88	4.94	5.00	5.06	5.11	5.15	5.20	5.24
	120	2.80	3.36	3.68	3.92	4.10	4.24	4.36	4.47	4.56	4.64	4.71	4.78	4.84	4.90	4.95	5.00	5.04	5.09	5.13
	∞	2.77	3.31	3.63	3.86	4.03	4.17	4.29	4.39	4.47	4.55	4.62	4.68	4.74	4.80	4.85	4.89	4.93	4.97	5.01

Tukey Test Critical Values Table

$\alpha = 0.10$																				
		k																		
		2	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
v	**1**	8.93	13.44	16.36	18.49	20.15	21.51	22.64	23.62	24.48	25.24	25.92	26.54	27.10	27.62	28.10	28.54	28.96	29.35	29.71
	2	4.13	5.73	6.77	7.54	8.14	8.63	9.05	9.41	9.72	10.01	10.26	10.49	10.70	10.89	11.07	11.24	11.39	11.54	11.68
	3	3.33	4.47	5.20	5.74	6.16	6.51	6.81	7.06	7.29	7.49	7.67	7.83	7.98	8.12	8.25	8.37	8.48	8.58	8.68
	4	3.01	3.98	4.59	5.03	5.39	5.68	5.93	6.14	6.33	6.49	6.65	6.78	6.91	7.02	7.13	7.23	7.33	7.41	7.50
	5	2.85	3.72	4.26	4.66	4.98	5.24	5.46	5.65	5.82	5.97	6.10	6.22	6.34	6.44	6.54	6.63	6.71	6.79	6.86
	6	2.75	3.56	4.07	4.44	4.73	4.97	5.17	5.34	5.50	5.64	5.76	5.87	5.98	6.07	6.16	6.25	6.32	6.40	6.47
	7	2.68	3.45	3.93	4.28	4.55	4.78	4.97	5.14	5.28	5.41	5.53	5.64	5.74	5.83	5.91	5.99	6.06	6.13	6.19
	8	2.63	3.37	3.83	4.17	4.43	4.65	4.83	4.99	5.13	5.25	5.36	5.46	5.56	5.64	5.72	5.80	5.87	5.93	6.00
	9	2.59	3.32	3.76	4.08	4.34	4.54	4.72	4.87	5.01	5.13	5.23	5.33	5.42	5.51	5.58	5.66	5.72	5.79	5.85
	10	2.56	3.27	3.70	4.02	4.26	4.47	4.64	4.78	4.91	5.03	5.13	5.23	5.32	5.40	5.47	5.54	5.61	5.67	5.73
	11	2.54	3.23	3.66	3.96	4.20	4.40	4.57	4.71	4.84	4.95	5.05	5.15	5.23	5.31	5.38	5.45	5.51	5.57	5.63
	12	2.52	3.20	3.62	3.92	4.16	4.35	4.51	4.65	4.78	4.89	4.99	5.08	5.16	5.24	5.31	5.37	5.44	5.49	5.55
	13	2.50	3.18	3.59	3.88	4.12	4.30	4.46	4.60	4.72	4.83	4.93	5.02	5.10	5.18	5.25	5.31	5.37	5.43	5.48
	14	2.49	3.16	3.56	3.85	4.08	4.27	4.42	4.56	4.68	4.79	4.88	4.97	5.05	5.12	5.19	5.26	5.32	5.37	5.43
	15	2.48	3.14	3.54	3.83	4.05	4.23	4.39	4.52	4.64	4.75	4.84	4.93	5.01	5.08	5.15	5.21	5.27	5.32	5.38
	16	2.47	3.12	3.52	3.80	4.03	4.21	4.36	4.49	4.61	4.71	4.81	4.89	4.97	5.04	5.11	5.17	5.23	5.28	5.33
	17	2.46	3.11	3.50	3.78	4.00	4.18	4.33	4.46	4.58	4.68	4.77	4.86	4.93	5.01	5.07	5.13	5.19	5.24	5.30
	18	2.45	3.10	3.49	3.77	3.98	4.16	4.31	4.44	4.55	4.65	4.75	4.83	4.90	4.98	5.04	5.10	5.16	5.21	5.26
	19	2.45	3.09	3.47	3.75	3.97	4.14	4.29	4.42	4.53	4.63	4.72	4.80	4.88	4.95	5.01	5.07	5.13	5.18	5.23
	20	2.44	3.08	3.46	3.74	3.95	4.12	4.27	4.40	4.51	4.61	4.70	4.78	4.85	4.92	4.99	5.05	5.10	5.16	5.20
	24	2.42	3.05	3.42	3.69	3.90	4.07	4.21	4.34	4.44	4.54	4.63	4.71	4.78	4.85	4.91	4.97	5.02	5.07	5.12
	30	2.40	3.02	3.39	3.65	3.85	4.02	4.16	4.28	4.38	4.47	4.56	4.64	4.71	4.77	4.83	4.89	4.94	4.99	5.03
	40	2.38	2.99	3.35	3.60	3.80	3.96	4.10	4.21	4.32	4.41	4.49	4.56	4.63	4.69	4.75	4.81	4.86	4.90	4.95
	60	2.36	2.96	3.31	3.56	3.75	3.91	4.04	4.16	4.25	4.34	4.42	4.49	4.56	4.62	4.67	4.73	4.78	4.82	4.86
	120	2.34	2.93	3.28	3.52	3.71	3.86	3.99	4.10	4.19	4.28	4.35	4.42	4.48	4.54	4.60	4.65	4.69	4.74	4.78
	∞	2.33	2.90	3.24	3.48	3.66	3.81	3.93	4.04	4.13	4.21	4.28	4.35	4.41	4.47	4.52	4.57	4.61	4.65	4.69